Martingale Pricing, Hedging, and No-Arbitrage

From martingale prices to replication and market completeness

Lecture 7

One question drives the whole lecture

The question

How do self-financing gains determine prices and hedges, and when can we hedge every claim?

1. Recall the pricing measure built in Lecture 6.
2. Derive martingale valuation from self-financing gains.
3. Replicate a dividend-paying call, then change the numeraire.
4. Recover hedges and test whether two Brownian risks are spanned.
5. Apply the framework to Asian claims and no-arbitrage; preview American exercise.

The Risk-Neutral Starting Point

Lecture 6 built the measure; Lecture 7 uses it

$$\mathbb{P} \xrightarrow[\text{Girsanov}]{\text{change weights}} \mathbb{Q} \xrightarrow{\text{discount}} \text{traded martingales}$$

Today

We start from an equivalent martingale measure $\mathbb{Q}$ and ask what it implies for prices, hedges, and market structure.

Standing caution

A stochastic integral becomes a pricing martingale only after the required integrability or admissibility check.

The money-market account fixes our pricing units

For now, use the money-market account

$$B_t^{(0)} = e^{\int_0^t r_s \, \mathrm{d}s}.$$

Express each traded price in these units: $\bar{S}_t^i = S_t^i / B_t^{(0)}$.

Equivalent martingale measure

$\mathbb{Q} \sim \mathbb{P}$, and the discounted traded gains are $\mathbb{Q}$-martingales when the needed integrability holds.

Later, a different numeraire will change both the units and the measure.

Why the discounted stock is a local martingale under $\mathbb{Q}$

For a non-dividend-paying stock under the risk-neutral measure,

$$\mathrm{d}S_t = r_t S_t \,\mathrm{d}t + \sigma_t S_t \,\mathrm{d}W_t^{\mathbb{Q}}, \qquad D_t = \exp\left(-\int_0^t r_u \,\mathrm{d}u\right).$$

The product rule gives

$$\mathrm{d}(D_t S_t) = D_t \sigma_t S_t \,\mathrm{d}W_t^{\mathbb{Q}}, \qquad D_t S_t = S_0 + \int_0^t D_u \sigma_u S_u \,\mathrm{d}W_u^{\mathbb{Q}}.$$

The only fact needed here

A well-defined stochastic integral with respect to Brownian motion is a local martingale. Extra integrability is needed to call it a true martingale.

Self-Financing Leads to Martingale Prices

Discounting removes the financing account

For now, assume the risky assets pay no dividends. Let

$$B_t^{(0)} = \exp\left(\int_0^t r_u \, \mathrm{d}u\right), \qquad D_t = \frac{1}{B_t^{(0)}}.$$

For a self-financing wealth process

$$X_t = \phi_t^0 B_t^{(0)} + \sum_{i=1}^m \phi_t^i S_t^i,$$

the discounted value is

$$\bar{X}_t = D_t X_t = \phi_t^0 + \sum_{i=1}^m \phi_t^i \bar{S}_t^i.$$

Self-financing becomes a discounted stochastic integral

The product rule gives

$$\begin{aligned} \mathrm{d}(D_t X_t) &= D_t \,\mathrm{d}X_t + X_t \,\mathrm{d}D_t \\ &= D_t \Delta_t \,\mathrm{d}S_t + r_t D_t \beta_t B_t^{(0)} \,\mathrm{d}t - r_t D_t X_t \,\mathrm{d}t \\ &= \Delta_t \big(D_t \,\mathrm{d}S_t - r_t D_t S_t \,\mathrm{d}t \big). \end{aligned}$$

Therefore

$$\boxed{\mathrm{d}\bar{X}_t = \Delta_t \,\mathrm{d}\bar{S}_t.}$$

Meaning

After discounting, the bank account disappears from the gain equation.

Replication becomes a martingale identity

Let a self-financing strategy replicate H at T, with wealth V_t. Under the required admissibility and integrability conditions,

$$\bar{V}_t = \frac{V_t}{B_t^{(0)}}$$

is a $\mathbb{Q}$-martingale and

$$\bar{V}_T = \frac{H}{B_T^{(0)}}.$$

Therefore

$$\boxed{V_t = B_t^{(0)} \mathbb{E}^{\mathbb{Q}} \left[\frac{H}{B_T^{(0)}} \middle| \mathcal{F}_t \right].}$$

Check bond and stock payoffs

Assume constant r and no dividends.

One unit of currency

For $H = 1$,

$$V_t = e^{-r(T-t)}.$$

One share of stock

For $H = S_T$,

$$V_t = S_t,$$

because $e^{-rt}S_t$ is a $\mathbb{Q}$-martingale.

YOUR TURN Verify both identities directly from the pricing formula.

Constant rates give the familiar pricing formula

If r is constant and $H = g(S_T)$ is replicable, then

$$\boxed{V_t = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[g(S_T) \mid \mathcal{F}_t].}$$

YOUR TURN Where is the physical drift μ in this formula?

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YOUR TURN Where is the physical drift μ in this formula?

It has been absorbed into the change from $\mathbb{P}$ to $\mathbb{Q}$.

Under $\mathbb{Q}$, the conditional stock law is explicit

For constant r, σ,

$$\frac{\mathrm{d}S_u}{S_u} = r\,\mathrm{d}u + \sigma\,\mathrm{d}W_u^{\mathbb{Q}}.$$

Hence

$$S_T = S_t \exp\left(\left(r - \frac{1}{2}\sigma^2\right)(T-t) + \sigma(W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}})\right),$$

with $W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}} \sim \mathcal{N}(0, T-t)$ conditionally on $\mathcal{F}_t$.

A digital payoff separates probability from price

Let

$$H = \mathbf{1}_{\{S_T > K\}}.$$

Then

$$V_t = e^{-r(T-t)}\mathbb{Q}(S_T > K \mid \mathcal{F}_t).$$

YOUR TURN Is the option price itself a probability?

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YOUR TURN Is the option price itself a probability?

No. It is a discounted risk-neutral probability.

A stock-or-nothing payoff uses a weighted tail

Let $H = S_T \mathbf{1}_{\{S_T > K\}}$. Then

$$V_t = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{\{S_T > K\}} \mid \mathcal{F}_t].$$

YOUR TURN Why is this not $e^{-r(T-t)} S_t \mathbb{Q}(S_T > K \mid \mathcal{F}_t)$?

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S_T and the event are dependent. The stock-weighted tail produces the d_+ term in Black–Scholes.

The forward price follows from one martingale identity

For a forward paying $S_T - K$ on a stock without dividends,

$$\boxed{V_t = S_t - Ke^{-r(T-t)}.}$$

The zero-value delivery price solves $V_t = 0$:

$$\boxed{F_{t,T} = S_t e^{r(T-t)}.}$$

Replication check

Long one share and borrow $Ke^{-r(T-t)}$. The terminal value is $S_T - K$, exactly the forward payoff.

A call is a stock tail minus a cash tail

For $H = (S_T - K)^+$,

$$(S_T - K)^+ = S_T \mathbf{1}_{\{S_T > K\}} - K \mathbf{1}_{\{S_T > K\}}.$$

Therefore

$$C_t = e^{-r\tau} \mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{\{S_T > K\}} \mid \mathcal{F}_t] - K e^{-r\tau} \mathbb{Q}(S_T > K \mid \mathcal{F}_t).$$

YOUR TURN Which term becomes the d_+ term and which becomes the d_- term?

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The stock-weighted tail gives d_+; the cash-or-nothing tail gives d_-.

Dividend Yield and Risk-Neutral Pricing

Dividends force us to distinguish price from gain

If one share pays continuous yield q_t, its instantaneous total gain is

$$\boxed{\mathrm{d}G_t = \mathrm{d}S_t + q_t S_t \,\mathrm{d}t.}$$

Ex-dividend price

S_t is the market value of the share after separating paid dividends.

Gain process

G_t records both price changes and cash distributions.

Pricing condition

Discounted total gains, not merely the ex-dividend price, must have zero drift.

Replicate a call on a dividend-paying stock

Under $\mathbb{P}$, assume constant r, q, σ, with μ the ex-dividend drift:

$$\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^{\mathbb{P}}, \qquad \mathrm{d}B_t = rB_t\,\mathrm{d}t.$$

For $V_t = v(t, S_t)$, Itô's formula gives

$$\mathrm{d}v = (v_t + \mu S_t v_s + \tfrac{1}{2}\sigma^2 S_t^2 v_{ss})\,\mathrm{d}t + \sigma S_t v_s\,\mathrm{d}W_t^{\mathbb{P}}.$$

A self-financing portfolio with Δ_t shares receives dividends:

$$\boxed{\mathrm{d}V_t = \Delta_t(\,\mathrm{d}S_t + qS_t\,\mathrm{d}t) + r(V_t - \Delta_t S_t)\,\mathrm{d}t.}$$

Matching the two gains gives the dividend PDE

Brownian exposure fixes the stock holding: $\Delta_t = v_s(t, S_t)$. Matching the drifts under $\mathbb{P}$ gives

$$v_t + \mu s v_s + \tfrac{1}{2}\sigma^2 s^2 v_{ss} = \mu s v_s + q s v_s + r(v - s v_s).$$

The historical drift cancels, leaving the replication equation

$$\boxed{v_t + \tfrac{1}{2}\sigma^2 s^2 v_{ss} + (r-q) s v_s - r v = 0, \qquad v(T, s) = (s-K)^+.}$$

NO-ARBITRAGE The call costs $v(t, S_t)$: the stock and bank-account strategy replicate its terminal payoff.

A prepaid share explains every dividend adjustment

A claim delivering one share at T is worth

$$\boxed{S_t e^{-q(T-t)}}.$$

YOUR TURN How many shares should we buy at time t if all dividends are reinvested?

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YOUR TURN How many shares should we buy at time t if all dividends are reinvested?

Buy $e^{-q(T-t)}$ shares. Reinvestment grows the position at rate q, so it reaches one share at T.

$$F_{t,T} = S_t e^{(r-q)(T-t)}.$$

Recover the risk-neutral measure with a dividend yield

Under $\mathbb{P}$, let μ be the *ex-dividend price drift*:

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{P}}, \qquad \text{dividend flow} = qS_t\,\mathrm{d}t.$$

Choose $\lambda = (\mu + q - r)/\sigma$ and set

$$Z_T = e^{-\lambda W_T^{\mathbb{P}} - \lambda^2 T/2}, \qquad W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \lambda t.$$

YOUR TURN Substitute $\mathrm{d}W_t^{\mathbb{P}} = \mathrm{d}W_t^{\mathbb{Q}} - \lambda\,\mathrm{d}t$. What is the new ex-dividend drift?

Recover the risk-neutral measure with a dividend yield

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$$\boxed{\mathrm{d}S_t/S_t = (r - q)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}.}$$

Solve the replication PDE with two Gaussian tails

The PDE's Feynman–Kac solution, with $\tau = T - t$, is

$$v(t,s) = e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+ \mid S_t = s].$$

Here

$$S_T = s\exp\big((r - q - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}Z\big), \quad Z \sim \mathcal{N}(0,1).$$

Define

$$d_2^q = \frac{\log(s/K) + (r - q - \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \qquad d_1^q = d_2^q + \sigma\sqrt{\tau}.$$

The cash and stock tails are

$$\mathbb{Q}(S_T > K \mid S_t = s) = \Phi(d_2^q), \qquad e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{\{S_T > K\}} \mid S_t = s] = se^{-q\tau}\Phi(d_1^q).$$

Replication fixes the dividend-adjusted call price and hedge

Using $(S_T - K)^+ = S_T \mathbf{1}_{\{S_T > K\}} - K \mathbf{1}_{\{S_T > K\}}$,

$$\boxed{C_t = S_t e^{-q\tau} \Phi(d_1^q) - K e^{-r\tau} \Phi(d_2^q).}$$

Differentiating the solution gives the replicating stock holding

$$\boxed{\Delta_t = v_s(t, S_t) = e^{-q\tau} \Phi(d_1^q),} \qquad C_t - \Delta_t S_t = -K e^{-r\tau} \Phi(d_2^q).$$

NO-ARBITRAGE This value solves the replication PDE and has the call payoff at maturity; the stock and bank-account holdings therefore fix its no-arbitrage price.

Changing the Numeraire

A general numeraire changes both units and measure

Let $N_t > 0$ be a traded numeraire. Under its associated measure $\mathbb{Q}^N$, traded prices expressed in units of N are martingales:

$$\frac{X_t}{N_t} = \mathbb{E}^{\mathbb{Q}^N}\left[\frac{X_T}{N_T}\middle|\mathcal{F}_t\right].$$

Thus a replicable payoff H has value

$$\boxed{V_t = N_t \mathbb{E}^{\mathbb{Q}^N}\left[\frac{H}{N_T}\middle|\mathcal{F}_t\right].}$$

Why choose another numeraire?

A suitable unit can turn a complicated discounted payoff into a simpler conditional expectation.

The best numeraire can make a payoff trivial

Suppose $H = N_T Y_T$. Then

$$\frac{V_t}{N_t} = \mathbb{E}^{\mathbb{Q}^N}[Y_T \mid \mathcal{F}_t].$$

YOUR TURN What happens if $H = N_T$? What is the price without any calculation?

The best numeraire can make a payoff trivial

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$$Y_T = 1 \quad \Longrightarrow \quad \boxed{V_t = N_t.}$$

Pricing becomes transparent when the payoff is naturally expressed in the chosen unit.

From Prices to Hedges: Martingale Representation

Start from a terminal variable and its conditional expectation

Let $\xi \in L^2(\mathbb{Q})$ be a terminal random variable and define

$$\boxed{M_t := \mathbb{E}^{\mathbb{Q}}[\xi \mid \mathcal{F}_t].}$$

Then M is a square-integrable martingale, with $M_T = \xi$.

What is missing?

The conditional expectation identifies the martingale. How do we express its random increments as a stochastic integral?

The route

Terminal variable $\xi \longrightarrow$ martingale $M \longrightarrow$ Brownian coefficient $\psi \longrightarrow$ hedge holdings.

The MRT decomposes every Brownian martingale into shocks

Assume $(\mathcal{F}_t)$ is the augmented filtration of a d-dimensional $\mathbb{Q}$-Brownian motion $W^{\mathbb{Q}}$, and $\xi \in L^2(\mathbb{Q})$.

Martingale Representation Theorem

There exists a predictable process $\psi_t \in \mathbb{R}^d$ such that

$$M_t = M_0 + \int_0^t \psi_u^\top \, \mathrm{d}W_u^{\mathbb{Q}},$$

with $\mathbb{E}^{\mathbb{Q}}[\int_0^T \|\psi_u\|^2 \, \mathrm{d}u] < \infty$.

Interpretation

ψ_t is the martingale's instantaneous exposure to each Brownian risk.

Example: the representation of a linear terminal shock

Take $d = 1$ and $\xi = W_T^{\mathbb{Q}}$. Write

$$W_T^{\mathbb{Q}} = W_t^{\mathbb{Q}} + (W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}}).$$

The increment is independent of $\mathcal{F}_t$ and has mean zero, so

$$M_t = \mathbb{E}^{\mathbb{Q}}[W_T^{\mathbb{Q}} \mid \mathcal{F}_t] = W_t^{\mathbb{Q}}.$$

$$\boxed{M_t = M_0 + \int_0^t 1 \, \mathrm{d}W_u^{\mathbb{Q}},} \qquad \boxed{\psi_t = 1.}$$

Reading

The claim carries one unit of Brownian exposure at every time.

Example: a quadratic claim has state-dependent exposure

Still with $d = 1$, take $\xi = (W_T^{\mathbb{Q}})^2$. Expanding the square gives

$$M_t = (W_t^{\mathbb{Q}})^2 + 2W_t^{\mathbb{Q}} \underbrace{\mathbb{E}^{\mathbb{Q}}[W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}} \mid \mathcal{F}_t]}_{0} + \underbrace{\mathbb{E}^{\mathbb{Q}}[(W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}})^2 \mid \mathcal{F}_t]}_{T-t} = (W_t^{\mathbb{Q}})^2 + (T - t).$$

Itô's formula shows the cancellation:

$$\mathrm{d}M_t = 2W_t^{\mathbb{Q}}\,\mathrm{d}W_t^{\mathbb{Q}} + \mathrm{d}t - \mathrm{d}t = 2W_t^{\mathbb{Q}}\,\mathrm{d}W_t^{\mathbb{Q}}.$$

$$\boxed{M_t = T + \int_0^t 2W_u^{\mathbb{Q}}\,\mathrm{d}W_u^{\mathbb{Q}},} \qquad \boxed{\psi_t = 2W_t^{\mathbb{Q}}.}$$

The discounted stock has its own Brownian exposure

For the hedging application, work directly with discounted values. With one stock, one Brownian factor, and no dividends, write

$$\boxed{\mathrm{d}\bar{S}_t = \sigma_t \bar{S}_t \, \mathrm{d}W_t^{\mathbb{Q}}.}$$

The terminal variable is now $\xi = \bar{V}_T$, so $M_t = \bar{V}_t$. If a self-financing hedge holds Δ_t shares, then

$$\mathrm{d}\bar{V}_t = \Delta_t \, \mathrm{d}\bar{S}_t = \Delta_t \sigma_t \bar{S}_t \, \mathrm{d}W_t^{\mathbb{Q}}.$$

Matching Brownian coefficients reveals the hedge

The MRT gives

$$\mathrm{d}\bar{V}_t = \psi_t \, \mathrm{d}W_t^{\mathbb{Q}},$$

while self-financing gives

$$\mathrm{d}\bar{V}_t = \Delta_t \sigma_t \bar{S}_t \, \mathrm{d}W_t^{\mathbb{Q}}.$$

If $\sigma_t \bar{S}_t \neq 0$, matching the coefficients gives

$$\boxed{\Delta_t = \frac{\psi_t}{\sigma_t \bar{S}_t}.}$$

Central message

The martingale approach does not lose the hedge. It stores it inside ψ_t.

In a smooth Markov model the MRT hedge is the usual delta

Suppose $V_t = v(t, S_t)$. The Brownian part of the discounted value is

$$\mathrm{d}\bar{V}_t = v_s(t, S_t)\sigma_t \bar{S}_t \,\mathrm{d}W_t^{\mathbb{Q}}.$$

Thus the MRT coefficient is

$$\psi_t = \sigma_t \bar{S}_t v_s(t, S_t).$$

Substitution into the MRT hedge formula gives

$$\boxed{\Delta_t = \frac{\sigma_t \bar{S}_t v_s(t, S_t)}{\sigma_t \bar{S}_t} = v_s(t, S_t).}$$

NO-ARBITRAGE The PDE delta and the martingale-representation hedge are the same object.

Two Brownian Risks and Market Completeness

Begin again under the historical measure

Think of W^1 as an AI-sector shock and W^2 as a broad market shock. Under $\mathbb{P}$, model an AI stock S and a sector state Y by

$$\frac{\mathrm{d}S_t}{S_t} = \mu(Y_t)\,\mathrm{d}t + \sigma_1(Y_t)\,\mathrm{d}W_t^1 + \sigma_2\,\mathrm{d}W_t^2, \qquad \mathrm{d}Y_t = a(Y_t)\,\mathrm{d}t + b(Y_t)\,\mathrm{d}W_t^1.$$

Assume $\sigma_2 \neq 0$. The sector state affects stock volatility, but Y is not traded.

YOUR TURN We assumed W^1 and W^2 independent. Is that realistic or useful? What might change if sector and market shocks are correlated?

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YOUR TURN We assumed W^1 and W^2 independent. Is that realistic or useful? What might change if sector and market shocks are correlated?

Independence simplifies the calculation; observed sector and market shocks may be correlated.

One traded stock does not hedge every call exposure

For a call $H = (S_T - K)^+$, write $V_t = v(t, S_t, Y_t)$. Its two Brownian coefficients from Itô's formula are

$$\underbrace{\big(S_t\sigma_1(Y_t)v_s + b(Y_t)v_y,\ S_t\sigma_2 v_s\big)}_{\text{call exposure}}.$$

Holding Δ_t shares gives only

$$\underbrace{\big(\Delta_t S_t\sigma_1(Y_t),\ \Delta_t S_t\sigma_2\big)}_{\text{stock exposure}}.$$

Where replication fails

The market component forces $\Delta_t = v_s$, leaving the sector exposure $b(Y_t)v_y$ uncovered when it is nonzero.

Why the extra state is essential to this example

If S had constant volatility and the payoff depended only on S_T, then $v(t, S_t)$ would have $v_y = 0$: ordinary delta hedging would still replicate the call, even with two Brownian terms in S.

The actual obstruction

Here sector conditions change the call value beyond today's stock price. That produces a second, independent hedge exposure through v_y.

YOUR TURN What additional traded instrument could supply exposure to the AI-sector shock?

A second traded asset can span the missing shock

Suppose another traded asset R has its own drift and both exposures:

$$\frac{\mathrm{d}R_t}{R_t} = \nu(Y_t)\,\mathrm{d}t + \eta_1(Y_t)\,\mathrm{d}W_t^1 + \eta_2(Y_t)\,\mathrm{d}W_t^2.$$

The stock and R can match the call's stochastic terms when

$$\begin{pmatrix} S\sigma_1 & R\eta_1 \\ S\sigma_2 & R\eta_2 \end{pmatrix} \begin{pmatrix} \Delta^S \\ \Delta^R \end{pmatrix} = \begin{pmatrix} S\sigma_1 v_s + b v_y \\ S\sigma_2 v_s \end{pmatrix}.$$

Rank condition

The hedge is unique when $\sigma_1\eta_2 - \sigma_2\eta_1 \neq 0$: the two traded assets cover two independent shock directions.

This spanning property is market completeness

A market is *complete* when every suitable terminal payoff in the modeled filtration can be replicated by self-financing trading.

two Brownian risks + two independent traded exposures $\Longrightarrow$ complete market

The rank criterion applies under the usual admissibility and integrability conditions.

Full rank

Both risk directions are spanned; claims have unique replication prices.

Rank one

Some risks remain unhedged; no-arbitrage may leave a range of prices.

The martingale route reaches the same hedge

With full rank, solve for the market prices of risk λ_1, λ_2:

$$\begin{pmatrix} \sigma_1 & \sigma_2 \\ \eta_1 & \eta_2 \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} \mu - r \\ \nu - r \end{pmatrix}.$$

Assuming the change-of-measure density is valid, $\mathrm{d}W_t^{i,\mathbb{Q}} = \mathrm{d}W_t^{i,\mathbb{P}} + \lambda_i(t)\,\mathrm{d}t$, and both discounted asset prices are $\mathbb{Q}$-local martingales.

$$\boxed{V_t = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+ \mid \mathcal{F}_t].}$$

Martingale representation gives two shock coefficients; matching them to the two traded assets recovers (Δ^S, Δ^R).

With several assets, hedging becomes a linear system

Let m discounted traded assets be driven by d Brownian factors:

$$\mathrm{d}\bar{S}_t = \Gamma_t \, \mathrm{d}W_t^{\mathbb{Q}}, \qquad \Gamma_t \in \mathbb{R}^{m \times d}.$$

If

$$\mathrm{d}\bar{V}_t = \psi_t^\top \, \mathrm{d}W_t^{\mathbb{Q}}$$

and the hedge uses holdings $\phi_t \in \mathbb{R}^m$, then

$$\mathrm{d}\bar{V}_t = \phi_t^\top \Gamma_t \, \mathrm{d}W_t^{\mathbb{Q}}.$$

Hence

$$\boxed{\phi_t^\top \Gamma_t = \psi_t^\top.}$$

Unspanned risk creates several martingale measures

Suppose the stock is driven only by W^1:

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^1.$$

Girsanov must choose

$$\lambda = \frac{\mu - r}{\sigma}$$

for W^1, but it may shift the independent W^2 by any η.

$$Z_T^\eta = \exp\left(-\lambda W_T^1 - \eta W_T^2 - \tfrac{1}{2}(\lambda^2 + \eta^2)\,T\right).$$

Every η prices the traded stock consistently.

A nontraded claim exposes the price ambiguity

Consider

$$H = (W_T^2)^2.$$

Under $\mathbb{Q}^\eta$,

$$W_T^2 \sim \mathcal{N}(-\eta T, T),$$

so

$$\boxed{\mathbb{E}^{\mathbb{Q}^\eta}[H] = T + \eta^2 T^2.}$$

YOUR TURN Why do different expectation values not create an arbitrage?

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so

$$\boxed{\mathbb{E}^{\mathbb{Q}^\eta}[H] = T + \eta^2 T^2.}$$

YOUR TURN Why do different expectation values not create an arbitrage?

Because H is not replicable by assets driven only by W^1.

Rank connects Brownian risks, hedges, and completeness

For

$$\mathrm{d}\bar{S}_t = \Gamma_t \, \mathrm{d}W_t^{\mathbb{Q}},$$

completeness requires every $\psi_t \in \mathbb{R}^d$ to be attainable as $\phi_t^\top \Gamma_t$.

Linear-algebra criterion

The traded volatility matrix must span all Brownian directions:

$$\boxed{\operatorname{rank}(\Gamma_t) = d.}$$

This requires at least as many independent traded exposures as Brownian risks.

A redundant asset can create many hedges without incompleteness

Let

$$\Gamma = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad d = 1, \quad m = 2.$$

For claim exposure $\psi = 3$, hedge holdings satisfy

$$\phi_1 + 2\phi_2 = 3.$$

YOUR TURN Give two different hedges. Does nonuniqueness of holdings imply nonuniqueness of the claim price?

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YOUR TURN Give two different hedges. Does nonuniqueness of holdings imply nonuniqueness of the claim price?

Examples are $(3, 0)$ and $(1, 1)$. The risk is fully spanned; the second asset is redundant.

Completeness concerns payoffs, not portfolio labels

Complete market

Every admissible Brownian exposure is spanned. Replicable claims have a unique no-arbitrage price.

Incomplete market

Some exposures are unspanned. Several EMMs may agree on traded assets but disagree on nontraded claims.

Do not confuse

Many portfolio representations may reflect redundant assets; many pricing measures reflect missing risk-spanning assets.

Asian Options and State Augmentation

Markov structure simplifies notation, not the pricing principle

Markov payoff

$$H = g(S_T), \qquad V_t = v(t, S_t).$$

The expectation may lead to a PDE or a closed form.

Path-dependent payoff

$$H = G((S_u)_{0 \le u \le T}).$$

The same conditional expectation remains valid on the full filtration.

Key distinction

The pricing identity is general; the low-dimensional state representation is the special feature.

A complete market can still have path-dependent payoffs

Return to the complete one-factor Black–Scholes market. A fixed-strike arithmetic Asian call pays

$$H = \left(\frac{1}{T} \int_0^T S_u \, \mathrm{d}u - K \right)^+ .$$

Why S_T is insufficient

Two paths can end at the same stock price while having very different running averages and therefore different payoffs.

A running integral restores a Markov state

Define

$$\boxed{A_t = \int_0^t S_u \, \mathrm{d}u.}$$

Then

$$\mathrm{d}A_t = S_t \, \mathrm{d}t, \qquad H = \left(\frac{A_T}{T} - K \right)^+.$$

If S is Markov, the pair (S_t, A_t) is Markov.

NO-ARBITRAGE Path dependence often becomes ordinary state dependence after adding the right accumulated variable.

The pricing identity survives state enlargement unchanged

Under constant r,

$$\boxed{V_t = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}\left[\left(\frac{A_T}{T} - K\right)^+ \middle| \mathcal{F}_t\right].}$$

In a Markov specification,

$$V_t = v(t, S_t, A_t).$$

What changed?

Not the expectation formula. Only the state required to evaluate it.

The running average adds a transport term

Under $\mathbb{Q}$, $\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^{\mathbb{Q}}$ and $\mathrm{d}A_t = S_t\,\mathrm{d}t$. Itô's formula for $v(t, S_t, A_t)$ gives

$$\mathrm{d}v = \left(v_t + rs\,v_s + s\,v_a + \tfrac{1}{2}\sigma^2 s^2 v_{ss}\right)\,\mathrm{d}t + \sigma s\,v_s\,\mathrm{d}W_t^{\mathbb{Q}}.$$

Therefore

$$\boxed{v_t + rs\,v_s + s\,v_a + \tfrac{1}{2}\sigma^2 s^2 v_{ss} - rv = 0,}$$

with

$$v(T, s, a) = \left(\frac{a}{T} - K\right)^+.$$

Arithmetic and geometric averages behave differently

Arithmetic average

$$\frac{1}{T}\int_0^T S_u \,\mathrm{d}u.$$

Common in contracts, but generally no simple Black–Scholes closed form.

Geometric average

$$\exp\left(\frac{1}{T}\int_0^T \log S_u \,\mathrm{d}u\right).$$

Its logarithm is Gaussian in Black–Scholes, making it tractable.

Monte Carlo handles path dependence without a closed form

For each simulated path under $\mathbb{Q}$:

1. generate $S_{t_1}, \ldots, S_{t_n}$;
2. compute the path average;
3. evaluate the payoff;
4. discount and average across paths.

$$\widehat{V}_0 = e^{-rT} \frac{1}{M} \sum_{m=1}^{M} \left(\frac{1}{n} \sum_{k=1}^{n} S_{t_k}^{(m)} - K \right)^+ .$$

Strength

Simulation uses the martingale pricing identity directly, even when the PDE state is high-dimensional.

Synthesis: Choosing the Right Route

The PDE and martingale routes answer the same pricing problem

PDE route

- exploit a Markov state;
- apply Itô and replication;
- read delta as a derivative;
- solve a differential equation.

Martingale route

- choose a numeraire and EMM;
- price by conditional expectation;
- recover the hedge through MRT;
- handle path dependence naturally.

The payoff structure suggests the most efficient tool

Problem	Natural first route
Smooth European payoff in one state	PDE or closed-form expectation
Linear payoff such as a forward	replication / law of one price
Path-dependent Asian payoff	conditional expectation / Monte Carlo
American exercise feature	optimal stopping / Snell envelope
Unspanned factor	EMM range / incomplete-market analysis

The diagnostic map separates five different ideas

1. Identical payoff $\Rightarrow$ law of one price.
2. Untraded Brownian risk $\Rightarrow$ incompleteness.
3. Early exercise $\Rightarrow$ optimal stopping and Snell envelope.
4. Running average $\Rightarrow$ state augmentation.
5. Martingale to holdings $\Rightarrow$ MRT and exposure matching.

Common thread

No-arbitrage restricts prices; spanning determines whether those restrictions select one price and one attainable payoff.

Use eight checks for every martingale-pricing problem

1. What is the numeraire and discount factor?
2. Under which measure are discounted traded gains martingales?
3. Is the payoff replicable, or is some risk unspanned?
4. Is the relevant local martingale a true martingale?
5. What state variables summarize the payoff?
6. Does exercise introduce an optimization over stopping times?
7. Which Brownian exposures appear in the MRT coefficient?
8. Do traded assets span those exposures?

Central message

Martingale pricing computes values, martingale representation reveals hedges, and completeness decides whether no-arbitrage selects a unique answer.

Revision Exercises: Lectures 2–6

Revision 1: apply Itô's formula to a sine

Let $X_0 = 0$ and

$$\mathrm{d}X_t = (1 + X_t)\,\mathrm{d}t + (1 + t)\,\mathrm{d}W_t, \qquad Y_t = \sin(X_t).$$

YOUR TURN Find $\mathrm{d}Y_t$, identifying its drift and Brownian coefficient.

1. Compute f' and f'' for $f(x) = \sin x$.
2. Compute $\mathrm{d}[X]_t$ from the Brownian coefficient of X.
3. Apply Itô's formula. Which term would the ordinary chain rule miss?

Revision 1: correction, from derivatives to the SDE

For $f(x) = \sin x$, $f'(x) = \cos x$ and $f''(x) = -\sin x$. The Brownian coefficient gives $\mathrm{d}[X]_t = (1+t)^2\,\mathrm{d}t$.

$$\begin{aligned}\mathrm{d}Y_t &= f'(X_t)\,\mathrm{d}X_t + \tfrac{1}{2}f''(X_t)\,\mathrm{d}[X]_t \\ &= \cos X_t\big[(1+X_t)\,\mathrm{d}t + (1+t)\,\mathrm{d}W_t\big] - \tfrac{1}{2}\sin X_t(1+t)^2\,\mathrm{d}t.\end{aligned}$$

$$\boxed{\begin{aligned}\mathrm{d}Y_t = {} & \big[(1+X_t)\cos X_t - \tfrac{1}{2}(1+t)^2\sin X_t\big]\,\mathrm{d}t \\ & + (1+t)\cos X_t\,\mathrm{d}W_t.\end{aligned}}$$

The extra drift $-\frac{1}{2}(1+t)^2\sin X_t$ comes from quadratic variation.

Revision 2: remove a deterministic drift with Girsanov

Under $\mathbb{P}$, let $Y_0 = 0$ and

$$\mathrm{d}Y_t = (\alpha + \beta t)\,\mathrm{d}t + \mathrm{d}W_t.$$

YOUR TURN Construct a measure $\mathbb{Q} \sim \mathbb{P}$ on $\mathcal{F}_T$ under which Y is Brownian.

1. Choose the drift shift and write its stochastic exponential.
2. Verify Novikov's condition on the finite horizon $[0, T]$.
3. Identify $W^{\mathbb{Q}}$ and check that it equals Y.

Revision 2: correction, construct a valid density

Set $\theta_t = \alpha + \beta t$ and define

$$Z_t = \exp\left(- \int_0^t \theta_s \, \mathrm{d}W_s - \tfrac{1}{2} \int_0^t \theta_s^2 \, \mathrm{d}s \right).$$

Since θ is deterministic,

$$I_T := \int_0^T \theta_s^2 \, \mathrm{d}s = \alpha^2 T + \alpha\beta T^2 + \frac{\beta^2 T^3}{3} < \infty.$$

$$\mathbb{E}^{\mathbb{P}}\left[e^{\frac{1}{2}\int_0^T \theta_s^2 \, \mathrm{d}s} \right] = e^{I_T/2} < \infty.$$

Novikov gives $\mathbb{E}^{\mathbb{P}}[Z_T] = 1$. As $Z_T > 0$, it defines

$$\boxed{\left. \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} \right|_{\mathcal{F}_T} = Z_T, \qquad \mathbb{Q} \sim \mathbb{P}.}$$

Revision 2: correction, check the sign of the shift

Girsanov's theorem says that

$$\boxed{W_t^{\mathbb{Q}} = W_t + \int_0^t \theta_s \, \mathrm{d}s = W_t + \alpha t + \tfrac{1}{2}\beta t^2}$$

is Brownian under $\mathbb{Q}$. Hence $\mathrm{d}W_t = \mathrm{d}W_t^{\mathbb{Q}} - \theta_t \, \mathrm{d}t$, and

$$\mathrm{d}Y_t = \theta_t \, \mathrm{d}t + \mathrm{d}W_t = \theta_t \, \mathrm{d}t + \mathrm{d}W_t^{\mathbb{Q}} - \theta_t \, \mathrm{d}t = \mathrm{d}W_t^{\mathbb{Q}}.$$

Since $Y_0 = W_0^{\mathbb{Q}} = 0$, we have $Y_t = W_t^{\mathbb{Q}}$.

Distribution check

Under $\mathbb{P}$, $Y_t \sim \mathcal{N}(\alpha t + \frac{1}{2}\beta t^2, t)$. Under $\mathbb{Q}$, $Y_t \sim \mathcal{N}(0, t)$.

Revision 3: the maximum of Brownian motion with drift

Under $\mathbb{P}$, let W be Brownian and define

$$X_t = W_t + \mu t, \qquad M_T^\mu = \max_{0 \le s \le T} X_s, \qquad T > 0, \quad a > 0.$$

YOUR TURN Compute $\mathbb{P}(M_T^\mu \ge a)$ using Girsanov and the reflection principle.

1. Make X Brownian under a new measure $\mathbb{Q}$.
2. Reflect paths at their first visit to a.
3. Return to $\mathbb{P}$ and evaluate the Gaussian integrals.

Here Φ denotes the standard normal distribution function.

Revision 3: correction, remove the drift first

Define

$$\left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = \exp\left(-\mu W_T - \tfrac{1}{2}\mu^2 T\right).$$

The constant shift satisfies Novikov. Under $\mathbb{Q}$, $X_t = W_t + \mu t$ is a standard Brownian motion.

$$\left.\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}}\right|_{\mathcal{F}_T} = \exp\left(\mu X_T - \tfrac{1}{2}\mu^2 T\right).$$

Therefore the desired probability is

$$\boxed{\mathbb{P}(M_T^\mu \geq a) = \mathbb{E}^{\mathbb{Q}}\left[e^{\mu X_T - \mu^2 T/2}\mathbf{1}_{\{M_T^\mu \geq a\}}\right].}$$

The maximum is evaluated on the same paths under both measures.

Revision 3: correction, reflect the paths ending below a

Under $\mathbb{Q}$, X_T has density $\varphi_T(x) = (2\pi T)^{-1/2} e^{-x^2/(2T)}$.

- If $X_T \geq a$, the continuous path has necessarily visited a.
- If $X_T = x < a$ after visiting a, reflection sends its endpoint to $2a - x > a$, preserving Brownian probability.

$$\mathbb{Q}(M_T^\mu \geq a,\ X_T \in \mathrm{d}x) = \begin{cases} \varphi_T(x)\,\mathrm{d}x, & x \geq a, \\ \varphi_T(2a - x)\,\mathrm{d}x, & x < a. \end{cases}$$

Weighting these two parts by $e^{\mu x - \mu^2 T/2}$ gives

$$\mathbb{P}(M_T^\mu \geq a) = \underbrace{\int_a^\infty e^{\mu x - \mu^2 T/2} \varphi_T(x)\,\mathrm{d}x}_{I_1} + \underbrace{\int_{-\infty}^a e^{\mu x - \mu^2 T/2} \varphi_T(2a - x)\,\mathrm{d}x}_{I_2}.$$

Revision 3: correction, evaluate the two Gaussian tails

Completing the square gives $e^{\mu x - \mu^2 T/2}\varphi_T(x) = \varphi_T(x - \mu T)$, so

$$I_1 = \int_a^\infty \varphi_T(x - \mu T)\,\mathrm{d}x = \Phi\left(\frac{\mu T - a}{\sqrt{T}}\right).$$

For I_2, substitute $y = 2a - x$:

$$I_2 = e^{2\mu a}\int_a^\infty e^{-\mu y - \mu^2 T/2}\varphi_T(y)\,\mathrm{d}y = e^{2\mu a}\Phi\left(\frac{-\mu T - a}{\sqrt{T}}\right).$$

$$\boxed{\mathbb{P}(M_T^\mu \geq a) = \Phi\left(\frac{\mu T - a}{\sqrt{T}}\right) + e^{2\mu a}\Phi\left(\frac{-\mu T - a}{\sqrt{T}}\right).}$$

For $\mu = 0$, we recover $2[1 - \Phi(a/\sqrt{T})]$.

Revision 4: which random times are stopping times?

Use the Brownian filtration $(\mathcal{F}_t)$. Fix $T > 0$ and $a > 0$.

YOUR TURN For each random time, decide whether $\{\tau \le t\} \in \mathcal{F}_t$ for every t.

1. $\tau_a = \inf\{t \ge 0 : W_t \ge a\}$;
2. $T/2$;
3. $\eta = \inf\{t \ge T/2 : W_t = 0\}$;
4. $\rho_T = \sup\{0 \le t \le T : W_t = 0\}$;
5. $\gamma_T = \arg\max_{0 \le s \le T} W_s$.

The Brownian maximum has a unique maximizing time almost surely.

Revision 4: correction, the first three are stopping times

First visit to a. By continuity,

$$\{\tau_a \le t\} = \left\{ \max_{0 \le s \le t} W_s \ge a \right\} \in \mathcal{F}_t.$$

The deterministic time $T/2$. Its event is

$$\{T/2 \le t\} = \begin{cases} \varnothing, & t < T/2, \\ \Omega, & t \ge T/2. \end{cases}$$

First zero after $T/2$. Again by continuity,

$$\{\eta \le t\} = \begin{cases} \varnothing, & t < T/2, \\ \left\{ \min_{T/2 \le s \le t} |W_s| = 0 \right\}, & t \ge T/2. \end{cases}$$

Each event can be decided from the path observed up to t.

Revision 4: correction, the last zero needs future information

Fix $0 < t < T$. The event is

$$\{\rho_T \le t\} = \{W \text{ has no zero in } (t, T]\}.$$

Knowing $W_t \neq 0$ does not tell us whether a future return to zero occurs. By independent increments and the reflection principle,

$$\boxed{\mathbb{P}(\rho_T \le t \mid \mathcal{F}_t) = 2\Phi\left(\frac{|W_t|}{\sqrt{T-t}}\right) - 1.}$$

For a fixed $0 < t < T$, this lies strictly between 0 and 1 almost surely.

Why this proves nonmeasurability

If $\{\rho_T \le t\}$ belonged to $\mathcal{F}_t$, its conditional probability would be its indicator, hence either 0 or 1. Thus ρ_T is not a stopping time.

Revision 4: correction, the maximizing time also looks ahead

Fix $0 < t < T$ and let $m_t = \max_{0 \le s \le t} W_s$. Up to a null event,

$$\{\gamma_T \le t\} = \left\{ \max_{t \le s \le T} W_s \le m_t \right\}.$$

The past tells us m_t and W_t, but a future increment may exceed the remaining gap $m_t - W_t$. Reflection gives

$$\boxed{\mathbb{P}(\gamma_T \le t \mid \mathcal{F}_t) = 2\Phi\left(\frac{m_t - W_t}{\sqrt{T-t}}\right) - 1.}$$

The same measurability test

For a fixed $0 < t < T$, the gap is positive almost surely, and the displayed probability lies in $(0, 1)$. It cannot be the indicator of an $\mathcal{F}_t$-measurable event. Thus γ_T is not a stopping time.

Final recap: quadratic variation and covariation

For partitions $\pi = (t_k)$ of $[0, t]$, put $\Delta X_k = X_{t_{k+1}} - X_{t_k}$. As the mesh tends to zero,

$$\sum_k (\Delta X_k)^2 \xrightarrow{\mathbb{P}} [X]_t, \qquad \sum_k \Delta X_k \Delta Y_k \xrightarrow{\mathbb{P}} [X, Y]_t.$$

What survives for continuous Itô processes

If $\mathrm{d}X_t = a_t\,\mathrm{d}t + \sum_i b_t^i\,\mathrm{d}W_t^i$ and $\mathrm{d}Y_t = c_t\,\mathrm{d}t + \sum_i d_t^i\,\mathrm{d}W_t^i$, with independent Brownian motions, then

$$\mathrm{d}[X]_t = \sum_i (b_t^i)^2\,\mathrm{d}t, \qquad \mathrm{d}[X, Y]_t = \sum_i b_t^i d_t^i\,\mathrm{d}t.$$

In particular, $[W]_t = t$; finite-variation terms contribute zero.

Final recap: the algebra of stochastic differentials

$$(\mathrm{d}t)^2 = 0, \qquad \mathrm{d}t\ \mathrm{d}W_t^i = 0, \qquad \mathrm{d}W_t^i\ \mathrm{d}W_t^j = \delta_{ij}\,\mathrm{d}t \quad \text{(independent Brownian motions)}.$$

Read $\mathrm{d}X_t\ \mathrm{d}Y_t$ as $\mathrm{d}[X, Y]_t$. Thus

$$\boxed{\mathrm{d}(X_t Y_t) = X_t\,\mathrm{d}Y_t + Y_t\,\mathrm{d}X_t + \mathrm{d}X_t\ \mathrm{d}Y_t.}$$

YOUR TURN If $\mathrm{d}X_t = 2\,\mathrm{d}t + 2\,\mathrm{d}W_t^1 + \mathrm{d}W_t^2$ and $\mathrm{d}Y_t = -\,\mathrm{d}t - \mathrm{d}W_t^1 + 3\,\mathrm{d}W_t^2$, what is $\mathrm{d}X_t\ \mathrm{d}Y_t$?

Final correction: expand the product of the increments

All products containing $\mathrm{d}t$ vanish, leaving

$$\begin{aligned}\mathrm{d}X_t\,\mathrm{d}Y_t &= (2\,\mathrm{d}W_t^1 + \mathrm{d}W_t^2)(-\,\mathrm{d}W_t^1 + 3\,\mathrm{d}W_t^2)\\ &= -2(\mathrm{d}W_t^1)^2 + 6\,\mathrm{d}W_t^1\,\mathrm{d}W_t^2 - \mathrm{d}W_t^2\,\mathrm{d}W_t^1 + 3(\mathrm{d}W_t^2)^2\\ &= (-2+3)\,\mathrm{d}t = \boxed{\mathrm{d}t}.\end{aligned}$$

The mixed Brownian products vanish because W^1 and W^2 are independent. Applying the product rule now gives

$$\boxed{\begin{aligned}\mathrm{d}(X_tY_t) = (2Y_t - X_t + 1)\,\mathrm{d}t\\ + (2Y_t - X_t)\,\mathrm{d}W_t^1 + (Y_t + 3X_t)\,\mathrm{d}W_t^2.\end{aligned}}$$